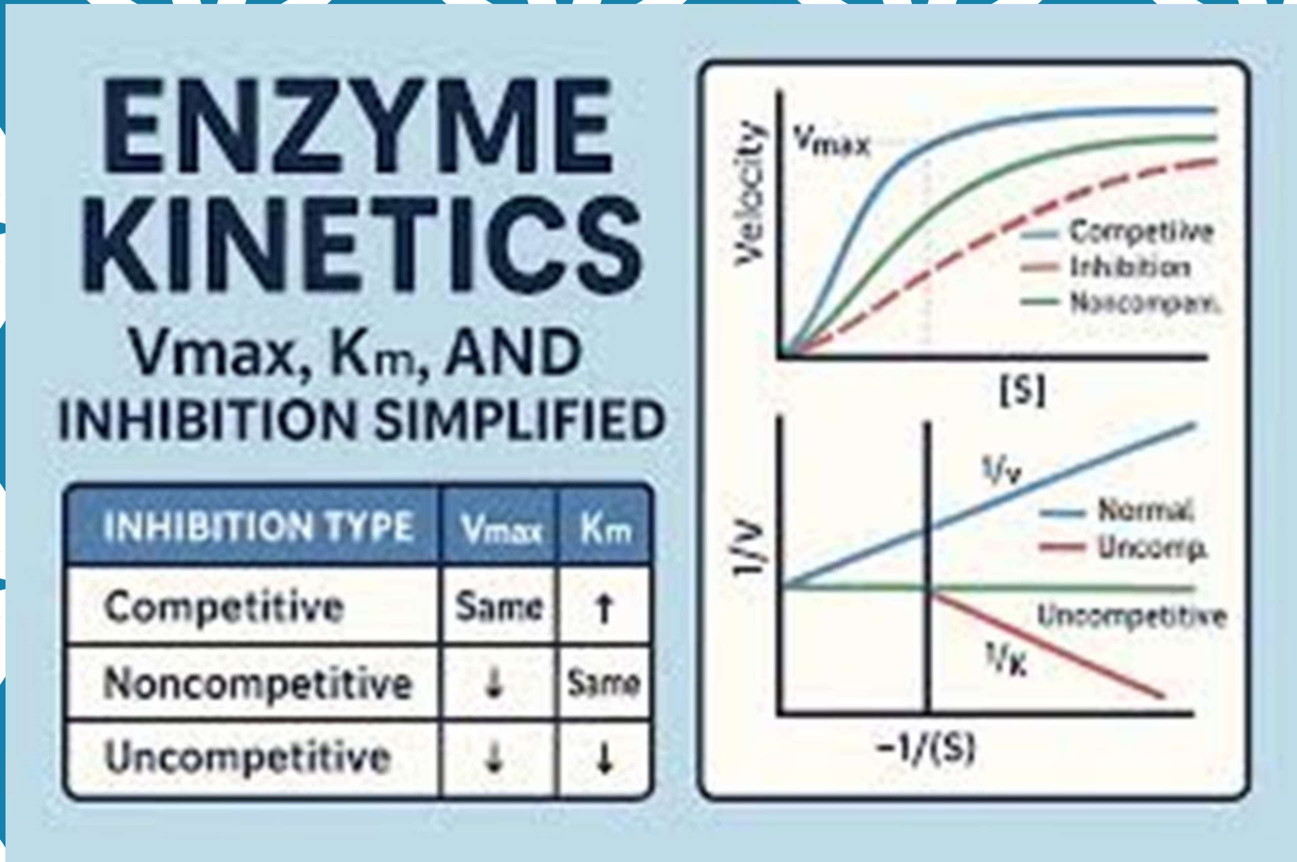

ENZYME KINETICS

DR WAJEEHA SHAMS ANSARI

Enzyme Kinetics

- Enzyme Kinetics – Quantitative measurement of the rates of enzyme catalyzed reactions

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- The systematic study of factors that affect these rates

- Enzyme kinetics began in 1902 when Adrina Brown reported an investigation of the rate of hydrolysis of sucrose as catalyzed by the yeast enzyme inveratase.

- Brown demonstrated – when sucrose concentration is much higher than that of the enzyme, reaction rate becomes independent of sucrose concentration

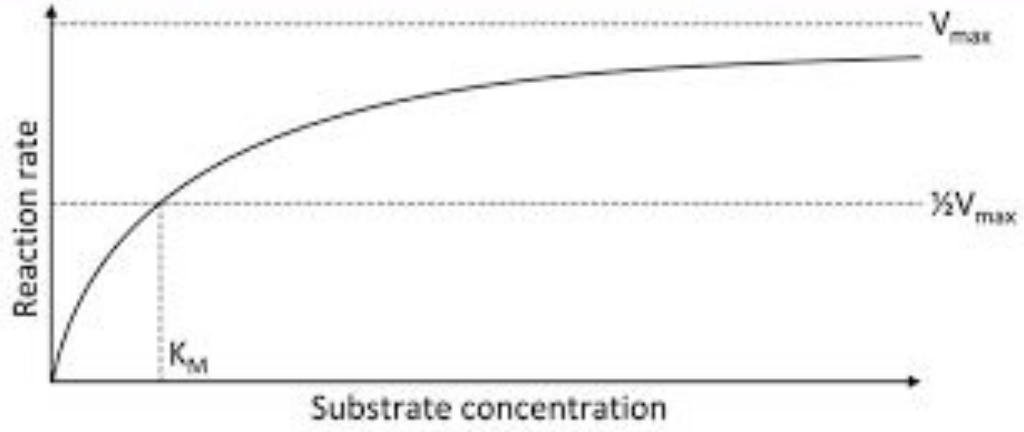
1. THE CORE VARIABLES

- When performing a kinetics practical, you are measuring the **Initial Velocity** (V_0) of a reaction at varying **Substrate Concentrations** ([S]).

- **V_{max}**: The maximum velocity at which the enzyme is saturated with substrate.

- **K_m (Michaelis Constant)**: The substrate concentration at which the velocity is half of V_{max}. It reflects the affinity of the enzyme for the substrate

The Michaelis–Menten Model (Kinetics)



$$V_o = \frac{V_{max}\,[S]}{K_m + [S]}$$

V0 = Initial velocity (moles/times)

[S] = substrate concentration (molar)

V_{max}= maximum velocity

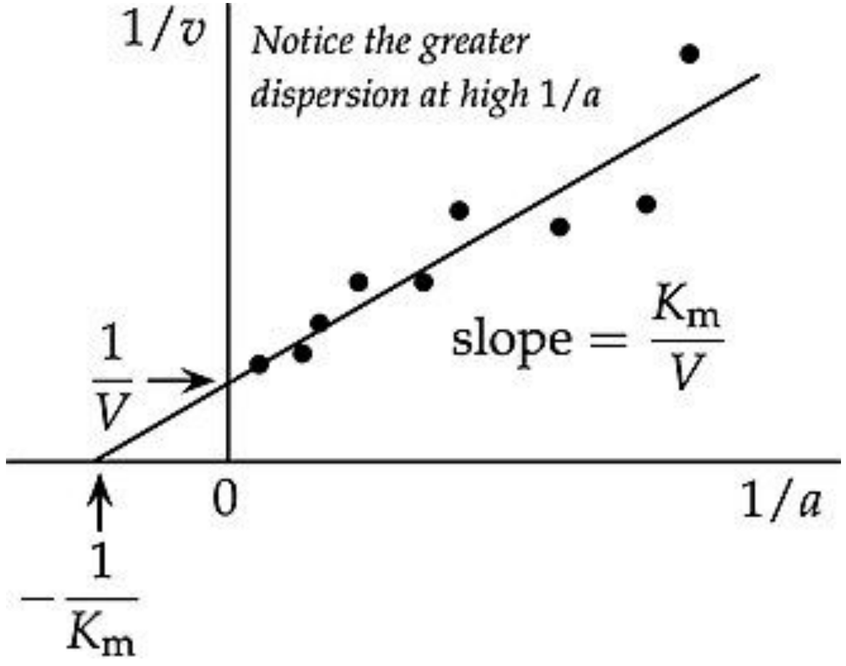
K_m = substrate concentration at half V_{max}

2. EXPERIMENTAL DATA ANALYSIS

•In a practical setting, plotting V_0 vs. [S] results in a hyperbolic curve, which is difficult to use for calculating exact values. **Lineweaver-Burk Plot** (Double-Reciprocal Plot) for linear analysis.

The Equation

$$\frac{1}{V_O} = \frac{K_m}{V_{max}\,(S)} + \frac{1}{V_{max}}$$



Feature	Kinetic Significance
y-intercept	$1/V_{max}$ (If the intercept moves up, V_{max} decreased)
x-intercept	$-1/K_m$ (If the intercept moves right, K_m increased)
Slope	K_m / V_{max}

3. PRACTICAL APPLICATION: ENZYME INHIBITION

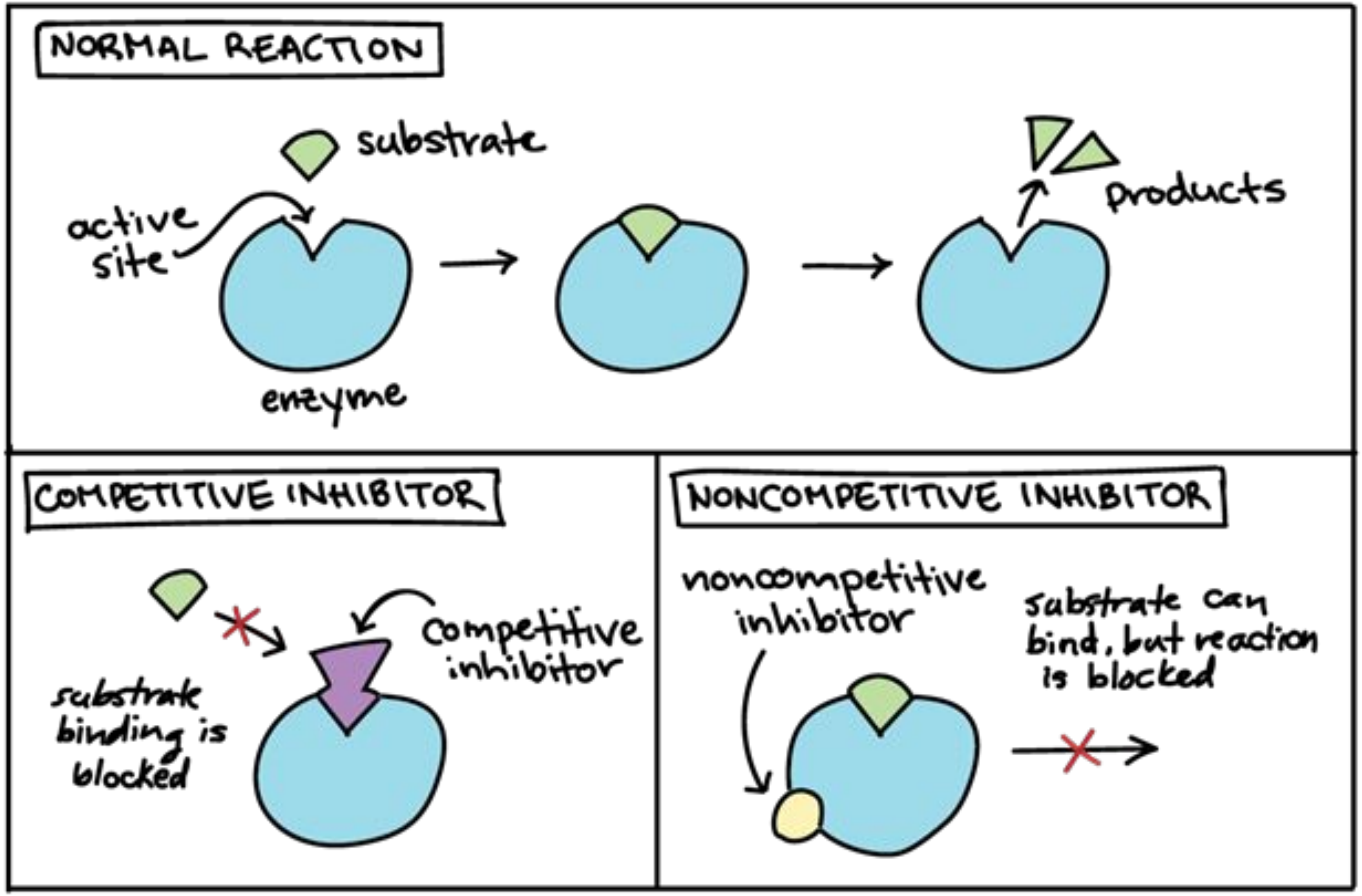
A common practical exercise involves determining how a drug (inhibitor) affects enzyme behavior.

Competitive Inhibition

- **Mechanism**: Inhibitor binds to the active site.
- **Effect**: V_{max} remains **unchanged**; K_m **increases** (requires more substrate to reach 1/2 V_{max}).
- **Graph**: The lines intersect at the y-axis.

Noncompetitive Inhibition

- **Mechanism**: Inhibitor binds to a different site (allosteric).
- **Effect**: V_{max} **decreases**; Km remains **unchanged**.
- **Graph**: The lines intersect at the x-axis.


NORMAL REACTION
substrate
active site
enzyme
products
COMPETITIVE INHIBITOR
competitive inhibitor
substrate binding is blocked
NONCOMPETITIVE INHIBITOR
noncompetitive inhibitor
substrate can bind, but reaction is blocked

4. CLINICAL CORRELATION

There are some clinical examples:

- **Statins**: Competitive inhibitors of HMG CoA reductase used to lower cholesterol.
- **Methanol Poisoning**: Ethanol acts as a competitive inhibitor for alcohol dehydrogenase, preventing the formation of toxic formaldehyde.
- **Ethylene Glycol**: Also treated with competitive inhibition using Fomepizole.


THANK YOU